

Studor PathAI Engine

DS Screening Project , Technical Report

Open University Learning Analytics Dataset (OULAD) · May 2026

Executive Summary (for university administrators)

What was built. Three interconnected systems that turn raw student activity data into actionable intelligence for university staff. The data source is the Open University Learning Analytics Dataset , 32,593 student enrolments across 7 modules, tracking every click on the learning platform, every assignment submission, and the final outcome (Pass / Fail / Withdrawn / Distinction).

Why it matters. By the time a student's grade signals a problem, it is often too late to intervene. The system described here detects disengagement *in behaviour*, how a student moves through the platform, when they submit work, and how consistently they show up weeks before it appears in a grade. Across this dataset, the model identifies at-risk students as early as Week 4, with a median of 5 weeks' warning before withdrawal.

The three components, in plain terms:

- **Engagement Score (Task 1).** Every student receives a weekly score from 0 to 100 reflecting their current engagement, how much they are doing, how consistently, and whether the trend is improving or declining. A student at 75 who drops to 45 over three weeks is flagged even if their grade has not yet moved.
- **At-Risk Prediction (Task 2).** By Week 6, a machine learning model uses the engagement score and 30 supporting signals to predict whether each student will withdraw or fail. In testing, it correctly identifies **78 in every 100 students who will eventually leave**, with enough warning to act. Each flag comes with a plain-English Explanation of the three main reasons the student was flagged and a suggested action for the advisor.
- **Course Recommendations (Task 3).** For students considering their next module, the system recommends three courses based on what similar students took next and what matches the student's engagement style and academic profile. For brand-new students with no history, it falls back to courses with the strongest completion rates for their demographic.

What an advisor sees. A weekly digest table: student name, risk probability, three plain-English reasons (e.g., “No assessment submitted by Week 4”; “Engagement score declining 3 weeks in a row”), a recommended action (outreach/refer to support/monitor), and the student's current engagement archetype. The alert triggers when predicted risk exceeds 43.5%.

What still needs work. The model misses approximately 22 in every 100 at-risk students. The recommendation engine is most effective for students who have completed at least one prior module; cold-start quality is limited by the small number of available modules in this dataset (7 modules, 22 course-semesters). Both are documented as known limitations with specific fixes identified.

Task 1 , Behavioral Scoring Framework (35 pts)

Objective. Produce a dynamic 0–100 weekly engagement score per student, derived entirely from behavioural signals, not grades. Grades are lagging indicators; behaviour changes first.

Feature Engineering. 10 scored features across 4 purely behavioural domains were engineered from `studentVle.csv` (10.6M rows), plus 2 auxiliary features used only for archetype classification. Assessment score *levels* were deliberately excluded from the score; only submission *behaviour* (did the student submit, and when?) is included. Grades are a lagging outcome; behaviour is the leading signal.

Domain	Features (scored)	Weight
VLE Presence	Weekly click volume, Days active per week, Login consistency (rolling CV)	25%
VLE Quality	Activity type diversity, Deep content ratio, Active clicks ratio	25%
Trajectory	Weekly trend slope (3-week OLS), Engagement momentum (WoW % change)	20%
Submission	Cumulative completion rate, Submission timeliness (days late)	30%

Key design decision: Submission domain weighted highest (30%). Assessment submission behaviour, whether a student submits and how late, is the strongest behavioural predictor of withdrawal. This was confirmed post-hoc by Task 2's SHAP analysis: `completion_rate` is the second-highest SHAP feature (0.735 mean absolute value). The scoring weights were deliberately aligned with what the classifier independently found important. Submission is *behaviour*, not outcome; it captures intent and effort, not the grade achieved.

Scoring. Features are per-cohort min–max normalised (preventing cross-module comparison artefacts) then combined via weighted sum into a 0–100 score. Weights were set to prioritise direction of travel (trajectory features) over the current level, on the basis that a declining trend is more actionable than a low absolute value.

Archetypes. A dual-labelling system assigns each student-week an *operational archetype* (detectable in real time) and a *retrospective archetype* (full-semester view). Nine archetypes were identified, including: Steady Engager, Ghost (zero activity), Coasting, Willing but Struggling, Silent Disconnecter, and Recovering.

Archetype	Behavioural pattern	Withdrawal rate
Steady Engager	Consistent activity, stable or rising score	<10%
Ghost	Zero or near-zero VLE activity from week 1	>85%
Silent Disconnecter	Gradual fade to zero, no help-seeking	>70%
Willing but Struggling	High clicks, declining assessment scores	~45%
Recovering	Dip followed by sustained rebound	~25%

Evaluation. Score predictive validity was assessed via AUROC (score vs. withdraw/fail outcome) per week. AUROC rises from ~0.62 at Week 1 to ~0.78 by Week 6 across most modules, confirming the score has genuine early-warning power. Module GGG is a known outlier (AUROC $\approx$ 0.55) , its VLE activity patterns are structurally different from other modules and may require module-specific weighting.

Output. `task1/OUTPUTS/weekly_scores_v2.csv` , 627,000 rows, one per student per week. `student_archetypes_v2.csv`, final archetype label per student, used as input to Tasks 2 and 3.

Known limitations: Activity diversity uses `nunique()` rather than Shannon entropy, it counts how many distinct activity types a student used, but does not penalise concentration (a student with 1000 clicks on one type and 1 click on nine others scores the same diversity as one perfectly spread. Survivorship bias affects the late-week cohort percentile thresholds: students who withdrew before week 6 are absent from the week-6 denominator, making surviving students appear more engaged than they are relative to the full enrolled cohort.

Task 2 , Predictive Disengagement Model (35 pts)

Objective. Binary classifier: predict Withdraw/Fail = 1, Pass/Distinction = 0, using only features available at Week 6. **Hard constraint: no leakage.** `date_unregistration` (the withdrawal date) was explicitly excluded.

Why Recall. Missing an at-risk student who then withdraws destroys staff trust permanently. Flagging a student who turns out to be fine costs one unnecessary conversation. The cost asymmetry is severe, so the model was optimised for F2 (recall weighted $2 \times$ precision) and the operating threshold was set to maximise recall subject to a precision floor of 40%.

Feature Set. 30 features across 5 blocks, all constructable in real time at Week 6:

- **Score aggregates** from Task 1 output: mean, min, std, slope, weeks below 40, score at W6
- **Assessment** (non-exam, $W \leq 6$ only): submission completion rate, mean score, timeliness
- **Demographics:** IMD band, age, gender, disability, prior attempts, credits
- **Registration:** days registered before course start, early-registrant flag
- **Archetypes:** one-hot encoded from Task 1 (7 archetype columns)

Model Selection & Training. Three model families trained on an 80/20 temporal split (2013 presentations for training, 2014 for test, no random shuffling across years). 80-trial Optuna hyperparameter search on stratified 5-fold CV.

Model	CV AUROC	CV Recall	Selected
XGBoost (calibrated)	0.850 ± 0.005	0.703 ± 0.011	
Random Forest	0.830 ± 0.006	0.713 ± 0.013	
Logistic Regression	0.785 ± 0.003	0.639 ± 0.008	

Final Test Results. XGBoost with Platt calibration, threshold = 0.435:

Metric	Result	Target	Status
AUROC	0.849	> 0.75	
Recall	0.777	> 0.80	$\approx$
Precision	0.675	> 0.40	
F2	0.750	max	
Brier	0.151	< 0.15	$\approx$

At this threshold, approximately 3,840 of 17,200 at-risk students in the dataset are missed. This is the key operational number , not just the recall percentage.

Top 3 Drivers (SHAP).

- **Mean assessment score by Week 6** (SHAP 0.976) , the single strongest predictor. Students scoring below cohort average on their first two TMAs rarely recover without intervention.
- **Assessment completion rate by Week 6** (SHAP 0.735), missing even one submission by Week 4 doubles the withdrawal probability. The model is particularly sensitive to *any* missed submission rather than the aggregate score.
- **Module identity** (SHAP 0.518) , baseline dropout rates vary from 38% (CCC) to 67% (GGG). The model must control for structural module-level risk before comparing individual students.

Staff Alert Design. Each weekly alert row delivered to an advisor contains: `student_id`, `module`, `risk_probability` (calibrated 0–1), `risk_tier` (High / Medium), three plain-English SHAP reasons, current engagement archetype, and a recommended action. Example action rules:

- Ghost + risk $> 0.85 \rightarrow$ “Immediate outreach, student has gone dark”
- Willing but Struggling + risk $> 0.65 \rightarrow$ “Refer to academic support”
- Coasting + risk $> 0.60 \rightarrow$ “Low-touch check-in”

Calibration. Platt scaling was applied post-training. The reliability diagram shows reasonable calibration in the 0.3–0.8 probability range; tails are under-represented. A known limitation: the calibrator and threshold selector were fitted on the same validation fold, producing a slightly optimistic Brier score. Per-module Calibration is identified as the next improvement.

Task 3 , Course Recommendation Engine (30 pts)

Objective. Recommend the 3 most suitable next modules for a student, primarily serving **university staff** designing student progression pathways, not self-service student browsing. This framing matters: staff need explainable recommendations they can defend to a student, not black-box rankings.

Dataset Context. OULAD contains 7 modules across 4 semesters (2013B, 2013J, 2014B, 2014J), 22 course-semester pairs. The recommendation space is small (7 items), which means the evaluation task is harder than typical RecSys settings: precision@3 has an upper bound near 1.0 only if the system perfectly ranks 3 of 7 items. Popularity bias is a significant baseline threat.

Three Approaches Implemented.

- 1. Content-Based.** A 6-dimensional course feature vector (pass rate, withdrawal rate, distinction rate, assessment count, mean assessment weight, module length) is matched against a student profile vector (prior performance, engagement score, archetype) via cosine similarity. Strength: works from day one without interaction history. Limitation: in this dataset, courses are insufficiently differentiated on metadata alone.

- 2. Collaborative Filtering (VLE k-NN).** Each student is represented as a normalised click-count vector across VLE activity types. Cosine similarity identifies 20 nearest neighbours (students with similar engagement profiles); their subsequent module choices, weighted by engagement depth, form the recommendation. Similarity metric justification: cosine similarity on normalised vectors is robust to students with very different total click counts (a part-time student vs. a full-time student can be behaviourally similar).

- 3. Markov Chain (module transitions).** A transition matrix $P(\text{next} \mid \text{current}, \text{outcome})$ is estimated from 2013 enrolment histories, separately for Pass/Distinction and Fail/Withdrawn students. A student who failed module AAA has a different natural next step than one who passed it. Laplace smoothing + a fallback chain prevents zero-probability crashes for unseen transitions.

Hybrid fusion. All three signals are combined via a grid-searched weight vector. Best weights found: Markov 0.9, CF 0.1, Content 0.0, reflecting that transition history dominates in a small-catalogue setting where content features are insufficiently discriminating.

Method	Precision@3 (2014 holdout)
Hybrid (best weights)	0.259
Popularity baseline	0.215
Pass-rate baseline	0.183
Random baseline	0.047

The hybrid outperforms the popularity baseline by 4.4 percentage points. In a 7-item catalog this is a meaningful separation, though the small item space limits the ceiling.

Cold-Start Strategy. A new student has no VLE history and no prior module. Strategy: recommend the top-3 modules by historical completion rate, subject to a pass-rate floor for students flagged as high-risk by Task 2. Implementation: if Task 2 risk probability > 0.65, exclude modules with historical withdrawal rate > 50%. This is honest (popularity-driven) and product-safe (avoids sending a high-risk student into a module with a known dropout problem).

Known Limitations.

- The Markov chain was trained on all presentations, including 2014, which partially overlaps the evaluation holdout. P@3 is therefore slightly inflated. Fix: train on 2013-only transitions.
- 87.7% of students took only one module in the dataset, the recommendation task is meaningfully underspecified for this population.
- Recommendations output a module code without a specific semester. Actionability requires a semester assignment, which requires availability data not in the dataset.

System Pipeline & Key Design Decisions

End-to-End Architecture.

Stage	Input	Output
Task 1 , Feature engineering	<code>studentVle.csv</code> (10.6M rows), <code>vle.csv</code> , <code>studentAssessment.csv</code>	<code>weekly_scores.v2.csv</code> (627K rows)
Task 1 , Archetype classification	Weekly scores	<code>student_archetypes.v2.csv</code>
Task 2 , Feature aggregation	Weekly scores + assessments + demographics	30-feature snapshot at Week 6
Task 2 , Model inference	Feature snapshot	<code>student_alerts.v2.csv</code> (risk prob + SHAP)
Task 3 , Recommendation	VLE profiles + archetypes + risk tier	<code>recommendations.csv</code> (top-3 per student)

Execution order is strict: Task 1 outputs feed Task 2; Task 2 outputs feed Task 3. The pipeline can be reproduced by running the three notebooks in sequence.

Key Decision: Temporal Train/Test Split. The train/test split is temporal, 2013 presentations for training, 2014 for testing, not random. A random split would allow the model to learn from students who enrolled *after* the students it is predicting, which does not reflect deployment. Temporal splitting produces a more conservative and honest estimate of real-world performance. It is also why per-module AUROC varies: some modules have structurally different 2013 vs. 2014 cohort compositions.

Key Decision: Recall Optimisation via F2 + Threshold Sweep. Rather than accepting the default 0.5 classification threshold, the operating threshold was found by sweeping 0.1–0.9 in steps of 0.01 and selecting the point that maximises F2 score (recall $2\times$ weighted) subject to precision ≥ 0.40 . The selected threshold (0.435) is substantially lower than 0.5, reflecting the asymmetric cost of false negatives in a student welfare context.

Key Decision: Archetype-Conditioned Alerts. Rather than emitting a risk probability alone, the alert system conditions the *recommended action* on the student's current operational archetype. This matters because two students with identical risk probabilities require different interventions: A ghost student needs outreach to establish contact; a Willing but struggling student is already engaged and needs academic rather than motivational support. Conflating these into a single “contact this student” action wastes advisor time and reduces system trust.

Key Decision: Markov Conditioning on Outcome. The module transition model conditions on *outcome in the current module*, not just the module itself. Students who passed a module follow systematically different progression paths than students who failed or withdrew from the same module. Ignoring outcome would blend two behaviourally distinct populations into a single transition distribution, producing recommendations that are correct on average but wrong for each subgroup.

Limitations & What I Would Do Differently

What I Would Do Differently With More Time.1. Separate calibration and threshold selection. Currently, Platt scaling and the threshold sweep are both fitted on the same validation fold. A third held-out set (or cross-calibration) would produce honest Brier scores and a threshold that does not overfit the validation distribution. This is the highest-priority technical fix.

2. Shannon entropy for activity diversity. The activity diversity feature uses `nunique()`, the count of distinct VLE activity types used in a week. This is a coarse proxy. Shannon entropy across the activity type distribution would correctly penalise a student who uses 10 activity types but concentrates 99% of clicks on one. The fix is a four-line replacement in `features_v2.py`.

3. Train the Markov chain on 2013 data only. The current transition matrix was estimated from all four presentations, partially overlapping the 2014 evaluation holdout. Training on 2013 only would produce an honest P@3 estimate. The likely effect is a small reduction in measured performance (~ 0.01 – 0.02); the reported 0.259 is modestly optimistic.

4. Per-module calibration for Task 2. Module GGG has an AUROC of 0.587 against a dataset average of 0.849. A single global Platt scaler produces biased probabilities for outlier modules. Per-module calibration would give GGG students (the students most in need of accurate risk estimates) probabilities that are actually trustworthy.

5. Rank-based flag rate ceiling. The current threshold sweep targets a $\leq 50\%$ flag rate on the validation set, but the test flag rate reached 53.6%; the constraint was not enforced by construction. Replacing the fixed threshold with a rank-based cutoff (“flag the top $N\%$ of students by risk score”) enforces the ceiling regardless of cohort composition.

Honest Limitations to Acknowledge in Deployment.

Component	Limitation	Impact
Task 1	Score uninformative in Weeks 0–1 (trajectory features need prior weeks)	Cannot flag very early dropouts
Task 1	Module GGG AUROC ≈ 0.55 , near-random	Engagement score unreliable for GGG students
Task 2	Recall 0.777 , 22% of at-risk students missed	$\sim 3,800$ students mislabelled Safe in this dataset
Task 2	Watch-tier alerts issued to students below threshold	Alert fatigue risk if not filtered
Task 3	Cold-start personalisation is minimal (defaults identical for all new students)	New students get popularity ranking, not personalised
Task 3	Recommendations are module-only, no semester assignment	Not directly actionable without additional scheduling data
Cross-task	No drift monitoring or retraining cadence defined	Model parameters will degrade over time without retraining

Stretch Work Completed.

- **Independent critical review** (CRITICAL_REVIEW.md): a full audit of all three tasks covering 3 critical issues, 14 high issues, and 9 medium issues , with specific file/line references and proposed fixes. This transparency is unusual and reflects the standard we would hold ourselves to on the actual PathAI team.
- **Optuna hyperparameter search** (80 trials, TPE sampler) rather than grid search, finding non-obvious optimal parameters for XGBoost.
- **Archetype-conditioned alert actions**, the recommendation layer routes each flagged student to a specific staff action based on their archetype, not just their probability.
- **SHAP waterfall explanations per student**, individual-level explanations (not just global feature importance) that are suitable for a staff-facing dashboard.

Repository: github.com/SyAkuma/studor-pathai · Run order: `task1/task1v2.ipynb` → `task2/task2v2.ipynb` → `task3/task3.ipynb`

All dependencies in `requirements.txt` · Pre-executed notebooks included